



Distribution of major ornamental fishes in the estuaries of Indian Sundarbans

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To cite this article:

Sana Ahmed, Ricardo Gobato, Sufia Zaman, and Abhijit Mitra. “Distribution of major ornamental fishes in the estuaries of Indian Sundarbans”, *Parana Journal of Science and Education*. Vol. 8, No. 6, **2022**, pp. 1-8. DOI: [tiny.cc/PJSE24476153v8i6p001-008](https://doi.org/10.24476/pjse.24476153v8i6p001-008)

Received: July 17, 2022; **Accepted:** July 31, 2022; **Published:** August 8, 2022.

Abstract

The estuaries of Indian Sundarbans are the hotspots of several categories of ornamental fishes that are primarily distributed as a function of salinity. Carried out a survey during November 2019 to January 2020 (postmonsoon period) at 10 stations distributed in the western and central sectors of Indian Sundarbans. Recorded 19 ornamental fish species during our survey, which exhibited significant distribution pattern based on the salinity of the ambient aquatic phase. It is interesting to note that the stations in the western sector of Indian Sundarbans namely, Harinbari (Stn. 1), Chemaguri (Stn. 2), Sagar South (Stn. 3), Lothian island (Stn. 4) and Prentice island (Stn. 5) showed the presence of all types of fishes, where as in the estuarine water surrounding the stations of central Indian Sundarbans namely, Canning (Stn. 6), Sajnekhali (Stn.7), Chotomollakhali (Stn. 8), Satjelia (Stn. 9) and Pakhiralaya (Stn. 10) most of the species are absent. The hypersalinity of the estuarine water might pose an adverse impact on the ornamental fish diversity in this sector of Indian Sundarbans.

Keywords: Indian Sundarbans, Estuary, Ornamental Fish Diversity and Salinity.

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1. Introduction

Mangrove forests in Indian Sundarbans with estuaries surrounding and criss-crossing it sustains a wide variety of fish fauna [1-3]. The brackish water of Indian Sundarbans estuaries is not uniformly saline *e.g.*, the estuaries in the western and eastern parts are hyposaline in nature, where as in the central sector the estuaries are hypersaline in nature [1, 4-9]. Because of such variations in salinity, the distribution of fishes exhibits a significant spatial variation probably due to their adaptive potential to different salinity. The estuarine water of Indian Sundarbans is taxonomically rich in ornamental fish diversity.

Ornamental fish species can provide an alternative livelihood to the island dwellers and to the local population of Indian Sundarbans who are mostly poverty stricken. Culture of these ornamental fishes will upgrade their economic profile as the ornamental fishes have wide internal and exports market. On this background was have made a survey in the post monsoon season (November 2019-January 2020) at 10 different stations distributed in the western and central sectors of Indian Sundarbans.

The Figure (1) represents three photos of the fish *Scatophagus argus*, for better illustration, of the fish of the studied estuaries. In Figure (1-C), there is a photo of a fossil of fish the scat family *Scatophagidae*. [10-12]

2. Materials and Methods

2.1 Collection and identification of samples

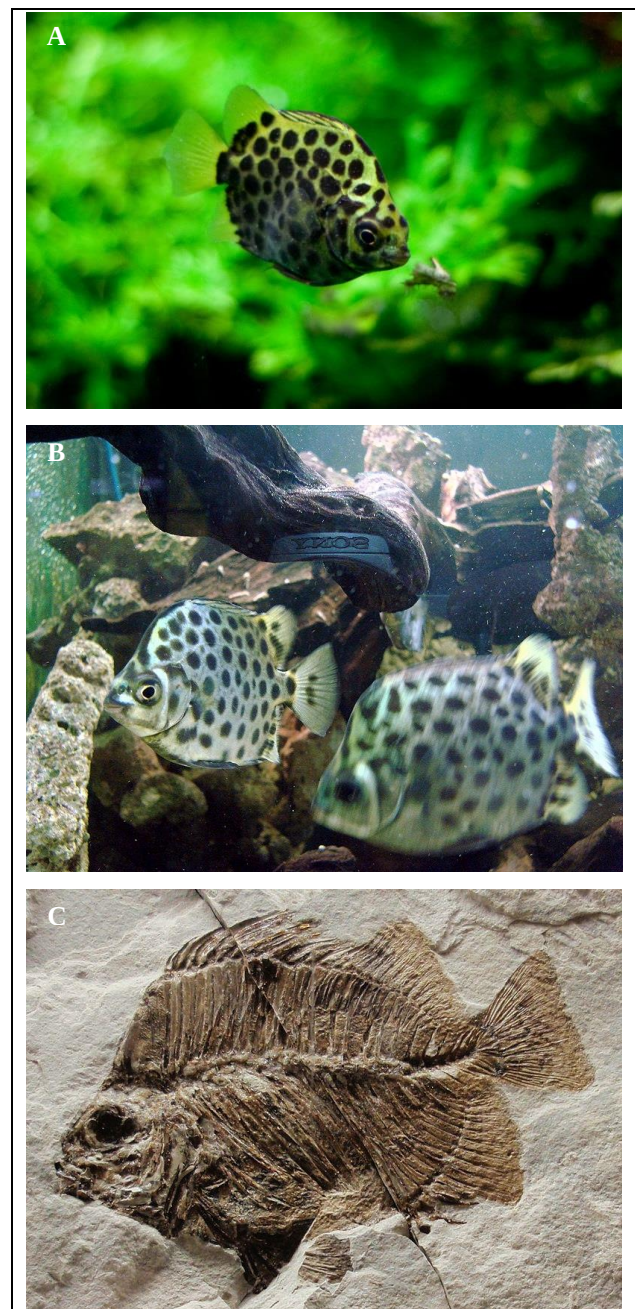
Bag nets, scoop nets and cast nets collected ornamental fish species between the periods November 2019 to January 2020. Some of these ornamental fishes were also collected from the local fisherman. Collections were done from 10 different stations distributed in the western and central Indian.

The *Scatophagus argus* the spotted scat, butterfly, “mia mia”, spotted butterfly or tiger scat, is a species of fish in the scat family *Scatophagidae*. It occurs in two basic color morphs which are called green scat and ruby or red scat. This fish is generally distributed around the Indo-Pacific region, to Japan, New Guinea, and southeastern Australia. They live in coastal muddy areas, including estuaries,

mangroves, harbors, and the lower courses of rivers. They are popular aquarium fish.

The *Scatophagidae*, the scats are a small family of ray-finned fishes in the order Perciformes. They are found in the Indo-Pacific region but one species was been introduced elsewhere.

Figure 1: Photo of fish *Scatophagus argus*.



Source: [10-12].



3. Monitoring the salinity

The surface water salinity was recorded during high tide condition from all the selected 10 stations by means of an optical refractometer (Atago, Japan), and cross checked in laboratory by employing Mohr- Kundsens method (after Strickland and Parsons, 1972). The correction factor was found out by titrating the silver nitrate solution against standard sea water (IAPO standard sea water service Charlottenlund, Slot Denmark, chlorinity 19.376 psu (Practical Salinity Unit).

4. Results

During our survey in the postmonsoon period covering 10 stations in the western and central sectors of Indian Sundarbans, was recorded 19 different species of ornamental fishes, Figure (3). Observed that the stations in the western sector (Stn. 1- Stn. 5), exhibited all varieties of fishes, but in the central sector (Stn. 6 - Stn.10) most of the species were either absent or rare.

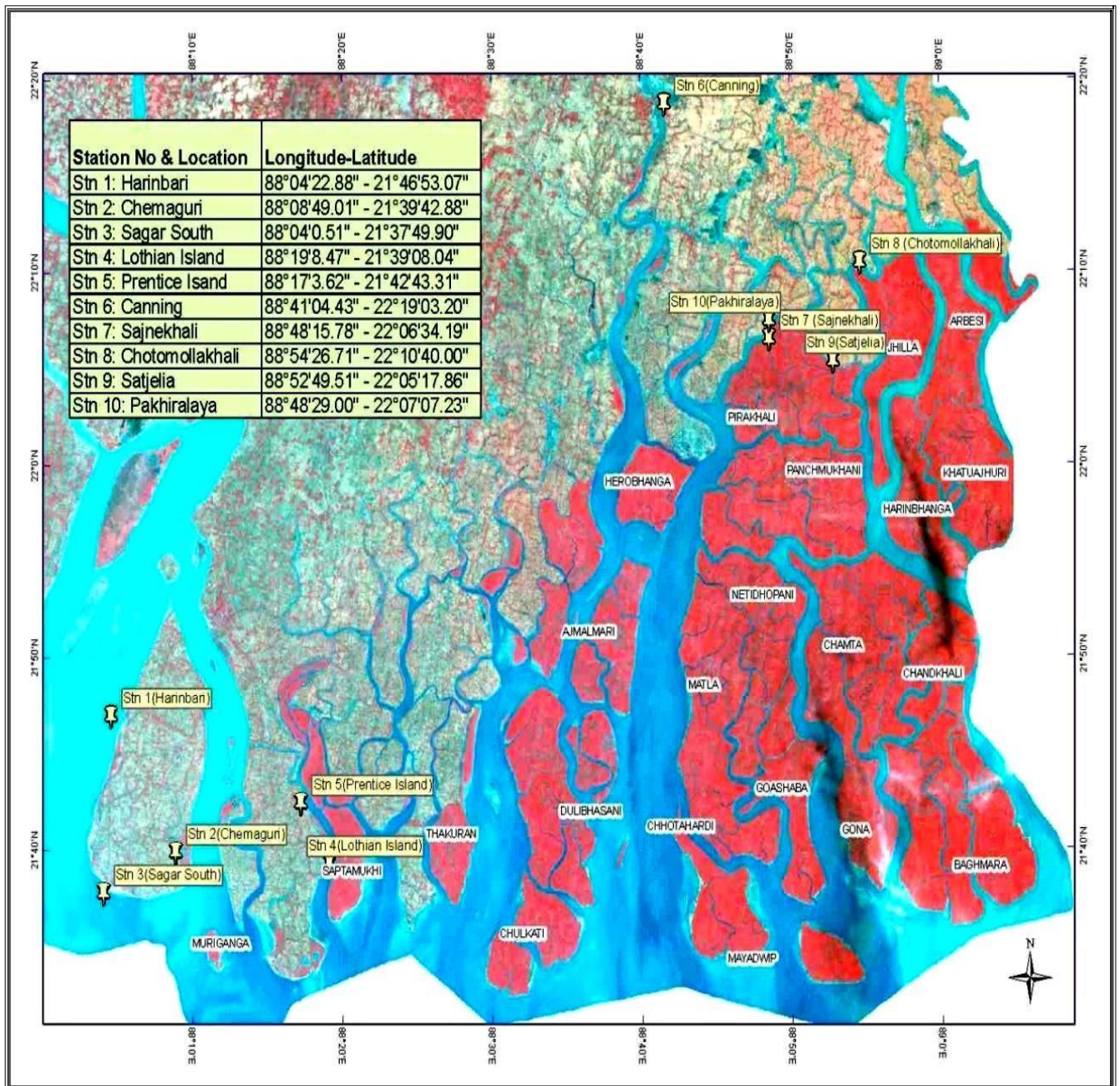
Table 1: Sampling stations with salient features.

Study site	Coordinates	Site Description
Harinbari (Stn. 1)	88°04'22.88" E 21°46'53.07" N	Situated in the western region of Indian Sundarbans almost in the middle of the Sagar Island; receives the water of the Hugli River.
Chemaguri (Stn. 2)	88°08'49.01" E 21°39'42.88" N	Situated on the south-eastern side of Sagar Island and receives the water of the Mooriganga River.
Sagar South (Stn. 3)	88°04' 0.51" E 21°37'49.90" N	Situated on the south-western part of the Sagar Island at the confluence of the River Hugli and the Bay of Bengal. Anthropogenically stressed zone due to presence of passenger jetties, fishing activities and Pilgrimage.
Lothian island (Stn. 4)	88°19'8.47"E 21°39'08.04" N	Situated east of Bakkhali island; a Wildlife sanctuary; faces the River Saptamukhi.
Prentice island (Stn. 5)	88°17'3.62"E 21°42'43.31" N	Situated north of Lothian island; receives the water of the Saptamukhi River.
Canning (Stn. 6)	88°41'04.43"E 22°19'03.20" N	Situated in the central part of the Indian Sundarbans and faces the mighty River Matla, a tide-fed river. Due to presence of fish landing stations, passenger jetties and busy market, the area is anthropogenically Stressed.
Sajnekhali (Stn. 7)	88°48'15.78"E 22°06'34.19" N	A Wildlife Sanctuary and a part of Sundarbans Tiger Reserve; adjacent to River Bidhya and Gomor. Tourism pressure is extremely high in this station particularly during postmonsoon.
Chotomollakhali (Stn. 8)	88°54'26.71" E 22°10'40.00" N	Situated in the upper portion of Central Indian Sundarbans adjacent to Jhila forest; receives the water of Rangabelia and Korankhali Rivers.
Satjelia (Stn. 9)	88°52'49.51"E 22°05'17.86" N	Situated adjacent to River Duttar in the upper region of Central Indian Sundarbans facing western part of the Jhilla forest.
Khiralaya (Stn. 10)	88°48'29.00"E 22°07'07.23" N	Situated adjacent to River Gomor; opposite To Sajnekhali Wild Life Sanctuary.

Source: Authors.



Figure 2: Map with station coordinates.

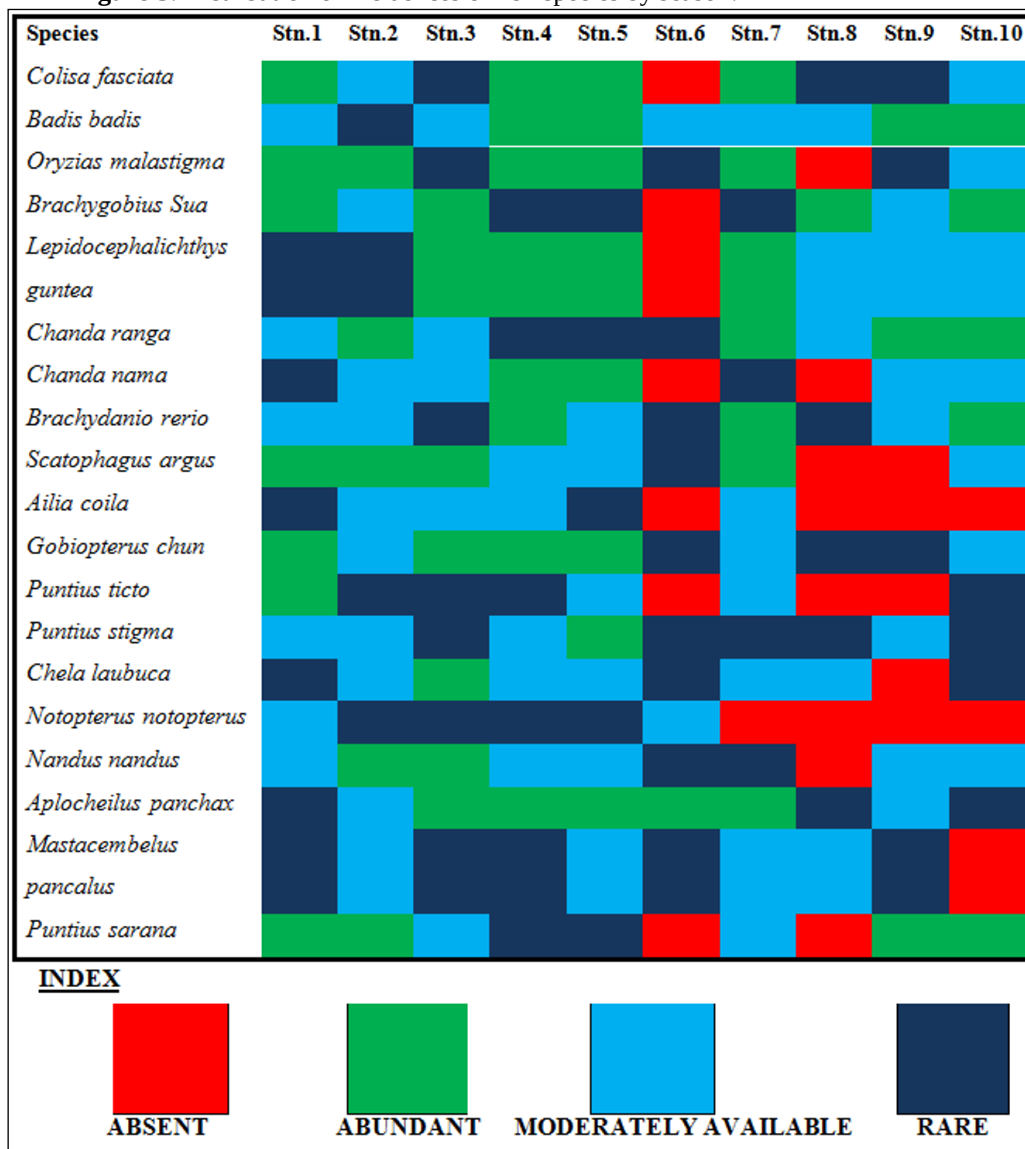


Source: Authors.

The Figure (2) shows the map with station coordinates.



Figure 3: Distribution of incidences of fish species by season.



Source: Authors.

Table 2: Surface Water salinity (in psu).

Sl No.	Name	Nov, 2019	Dec, 2019	Jan, 2020	Average	Sector
1.	Harinbari (Stn. 1)	7.33	9.49	9.58	8.8	Western sector water salinity = 19.54 psu
2.	Chemaguri (Stn. 2)	15.63	16.18	17.04	16.28	
3.	Sagar South (Stn. 3)	23.44	24.18	25.20	24.27	
4.	Lothian island (Stn. 4)	23.89	24.19	24.15	24.07	
5.	Prentice island (Stn. 5)	24.02	24.10	24.61	24.24	
Average salinity		18.86	19.62	20.12		
6.	Canning (Stn. 6)	11.66	15.82	18.21	15.23	Central sector water salinity = 24.56 psu
7.	Sajnekhali (Stn. 7)	24.95	26.38	29.05	26.79	
8.	Chotomollakhali (Stn. 8)	24.15	26.12	26.99	25.76	
9.	Satjelia (Stn. 9)	25.98	26.82	29.01	27.27	
10.	Pakhiralaya (Stn. 10)	26.85	27.12	29.34	27.77	
Average salinity		22.71	24.45	26.52		

Source: Authors.



5. Discussion

Ornamental fishes are noted for their beauty and diversity in color. The vibrant color of the fish species determines the cost of ornamental fish. It was been documented that marine and brackish water ornamental fishes are gradually getting momentum in the international market. Globally around 10-15 billion US dollars spin around the aquarium fish industry. The largest suppliers of marine ornamental fishes are Indonesia followed by Philippines. However, in India the trade is yet under-estimated and not much advancement has been done in the sectors of marine and brackish water ornamental fishery. The Indian Sundarbans with wide range of biodiversity is the natural breeding and survival ground of a variety of ornamental fishes, but their distribution is regulated by aquatic salinity as the adaptation of the ornamental fishes are primarily salinity-specific.

Was observed that salinity in the Matla estuary flowing through central Indian Sundarbans exhibits higher salinity than the Hooghly estuary in the western Indian Sundarbans, Table (2).

It is clearly visualized from the data sets that the central sector experiences a high saline aquatic sub-system (average value = 24.56 psu), which is the result of complete obstruction of the fresh water flow from the upstream region owing to Bidyadhari siltation since the late fifteenth century [1, 7, 13, 14]. The Matla estuary in the central Indian Sundarbans cannot be referred to as an ideal estuary as there is no head on discharge or dilution of the system with fresh water. Thus, Matla can be designated as a tidal channel, whose survival depends on the tidal flow from the Bay of Bengal.

The hyposaline environment of western Indian Sundarbans (average estuarine salinity = 19.54 psu) may be attributed to Farakka barrage discharge situated in the upstream region of Ganga- Bhagirathi-Hooghly River system. A long-term study conducted on salinity variation by several researchers in the Indian Sundarbans exhibits three distinct regimes in terms of salinity. The western sector is relatively low saline owing to fresh water discharge from the Farakka Barrage. Ten 10-year surveys (1999 to 2008) on water discharge from Farakka dam revealed an average discharge of $(3.7 \pm 1.15) \times 10^3 \text{ m}^3\text{s}^{-1}$. Higher discharge values were observed during the monsoon with an average of $(3.81 \pm 1.23) \times 10^3$

m^3s^{-1} , and the maximum of the order $4524 \text{ m}^3\text{s}^{-1}$ during freshet (September). Considerably lower discharge values were recorded during premonsoon with an average of $(1.18 \pm 0.08) \times 10^3 \text{ m}^3\text{s}^{-1}$, and the minimum of the order $846 \text{ m}^3\text{s}^{-1}$ during May. During postmonsoon discharge, values were moderate with an average of $(1.98 \pm 0.97) \times 10^3 \text{ m}^3\text{s}^{-1}$ [1].

The low diversity of ornamental fish in the central Indian Sundarbans is basically due to climate induced sea level rise and subsequent erosion, which causes the estuarine water highly turbid. Due to salinity intrusion into freshwater areas, many freshwater fish species are subjected to severe physiological stress and are unable to cope up with such extreme conditions [15-18]. These may have serious impacts on the distribution, feeding regimes, and breeding periodicity of the ornamental fish species.

6. Conclusions

The rising popularity of fish keeping as a hobby has increased the demand for ornamental fish worldwide, making them an important component of the global fish trade. Ninety per cent of the trade volume concentrates around tropical freshwater species, the majority of which is constituted by diverse wild-caught species. The wild collection of these species for the trade has become an arguable issue. Though ornamental fish trade could be an important contributor to local economies, the unmanaged wild collection can reduce the sustainability of the trade and lead to population decline of important fish species.

Despite being one among the top ten mega-diverse countries of the world in terms of fish diversity, India contributes less than 1% to the multi-billion-dollar global ornamental fish trade. With its enormous and diverse fish resources, India has immense scope to emerge as a potential candidate and strong competitor in this trade in the near future.

Indiscriminate harvesting of fish species from their natural habitat is regularly done by the rural people, which may lead to serious stock depletion. Through interviewing the local people, it is clear that the *Lepidocephalichthys guntea*, *Scatophagus argus*, *Gobiopertus chun*, *Oryzias malastigma*, *Brachyogobius sua*, *Ailia coilia*, *Mestaeembelus armatus*, *Macrognothus aculeatus*, *Mastacembelus pancalus* are facing a very



threatened condition, Table (1), but it is a fact that in earlier occasion these fishes were present in plenty. It is evident especially in case of *Puntius sarana* which is not available now but was common thirty years back. To overcome these problems, ornamental fishes can be reared and breed in captivity and vocational training could be provided to the fisher folk in rural areas for rearing and breeding of ornamental fishes, which have high demand in the market.

A good number of stakeholders are already involved in ornamental fish farming of coastal Bay of Bengal, but there is lack of proper marketing channel in this trade. In most of the cases, the rural women and children collect the fishes from the natural habitats. The exporters buy these fishes from the farmers/collectors, against nominal price and export them to the global fish market. To make this industry sustainable, a strong supply chain is required backed up with total quality management.

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